

LINDBERGH'S NONSTOP NEW YORK-TO-PARIS FLIGHT



"A SLIM, TALL, BASHFUL, SMILING American boy is somewhere over the Atlantic ocean, where no lone human being has ever ventured before... If he is lost, it will be the most universally regretted loss we ever had," wrote humorist Will Rogers on May 20, 1927. The "smiling American boy" was 25-year-old Charles Lindbergh, and Rogers' anxious concern for his fate was shared by millions in America and Europe. Lindbergh was the latest pilot to attempt the first nonstop flight between New York and Paris, a contest that had already cost the lives of six airmen. He intended to fly the 3,600 miles (5,760km) alone in a single-engined monoplane. Lindbergh's quiet courage and good looks had won him an enthusiastic following, but few people rated his chances.

SPIRIT OF ST. LOUIS

The Spirit of St. Louis' cramped cockpit, squeezed in behind the massive fuel tank, had no front view. Lindbergh had to turn the airplane to see ahead. He had bought the custom-built Ryan NYP with its 223hp Wright Whirlwind engine for \$10,580, using money raised from St. Louis businessmen.



Lindbergh's Ryan NYP was carrying 450 gallons (2000 liters) of fuel – a winged gasoline tank. He had jettisoned every luxury to lighten the airplane's load, carrying no radio and no sextant for navigation by the stars. His entire provisions for the journey were five sandwiches and two canteens of water. Yet still he did not know if the fuel-laden aircraft would get off the ground. Roosevelt Field was sodden and muddy and there was no headwind to aid takeoff. As the monoplane accelerated through mud and puddles, it twice rose and bumped back to the ground before at last achieving sustained lift and rising clear of the telegraph wires at the end of the runway.

Once in the air, Lindbergh faced two crucial problems. One was navigation. He had to find his way by dead reckoning, using a clock and an airspeed indicator to measure his distance traveled, and two compasses to plot his direction. Once he was over the ocean he would have no visual reference to check the accuracy of his calculation. His other problem was tiredness. He had not slept the night before the flight and faced a further day and a half without rest. By the time he left North America behind, crossing the coast of Newfoundland, he had already been in the air for 11 hours and was racked by fatigue. The 15-hour ocean crossing was a

FLYING LOW

The main difficulty that Lindbergh faced was staying awake for the 33½ hours his transatlantic flight took. This photograph was taken during his US tour, undertaken on his return home.

severe test of the flying skills and the instinct for survival that Lindbergh had developed as an airmail pilot. He had to fly for two hours in total darkness between sunset and moonrise; he flew through storm clouds and freezing cold that iced his wings; after daybreak he ran into thick fog. Few pilots could have managed to fly for so long in such poor visibility with the minimal instruments that Lindbergh possessed – essentially a turn-and-bank indicator and an altimeter. The routines of the flight helped

"Here, all around me, is the Atlantic – its expanse, its depth, its power... If my plane can stay aloft, if its engine can keep running, then so can I."

CHARLES LINDBERGH

keep him awake: regularly switching from one gas tank to another, noting the readings from his instruments in his log. So did the instability of his aircraft, which was definitely not designed to fly itself and would soon give the pilot a jolt if he nodded off at the controls.

By the morning of May 21, Lindbergh was hallucinating, creating mirages of land out of shapes in the fog. But first the sight of fishing boats and then a rugged coastline told him that he had made it across the ocean. Remarkably he was almost exactly on course, crossing the Irish coast

